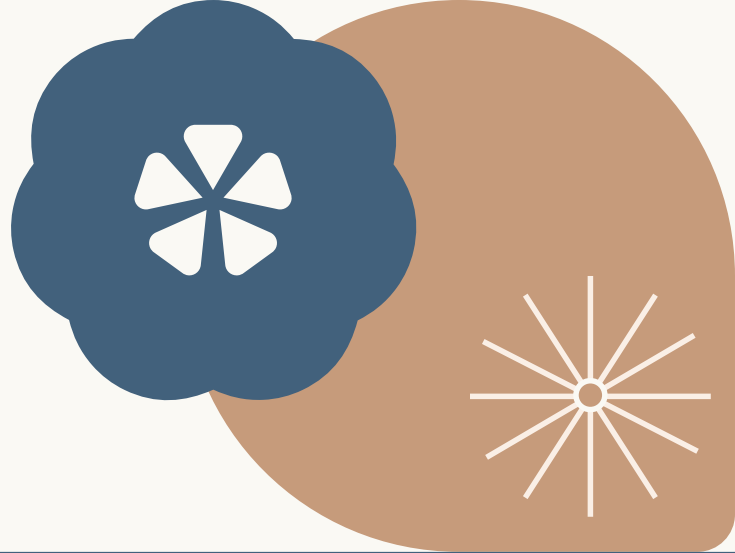


IMPACT OF FAMILY STRUCTURE & EDUCATION ON THE GENDER WAGE GAP IN MANAGEMENT ROLES

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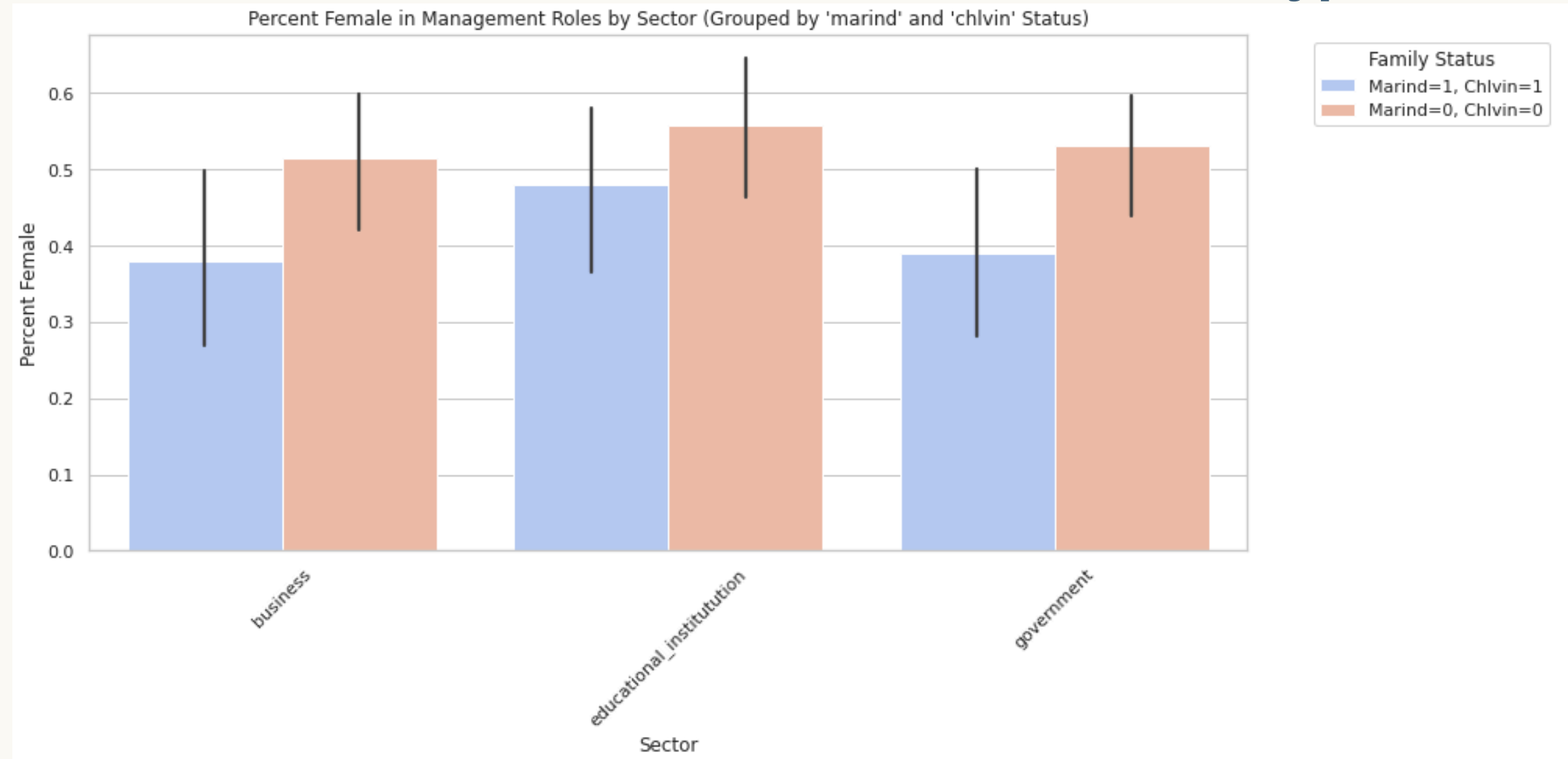


Project Overview



Research Question	What is the impact of family structure and education on the gender wage gap for individuals in management positions?
Hypothesis 1 (Chi Squared)	Null: Traditional family structures (partnered with children and married) do not significantly impact the gender wage gap and gender representation gap in management careers. Alternative: Traditional family structures (partnered with children and married) have a significant impact on the gender wage gap or gender representation gap in management careers.
Hypothesis 2 (Mediation Analysis)	Null: Education levels do not mediate the gender wage gap in management careers in the business sector for individuals who are partnered with children. Alternative: Education levels mediate the gender wage gap in management careers in the business sector for individuals who are partnered with children.

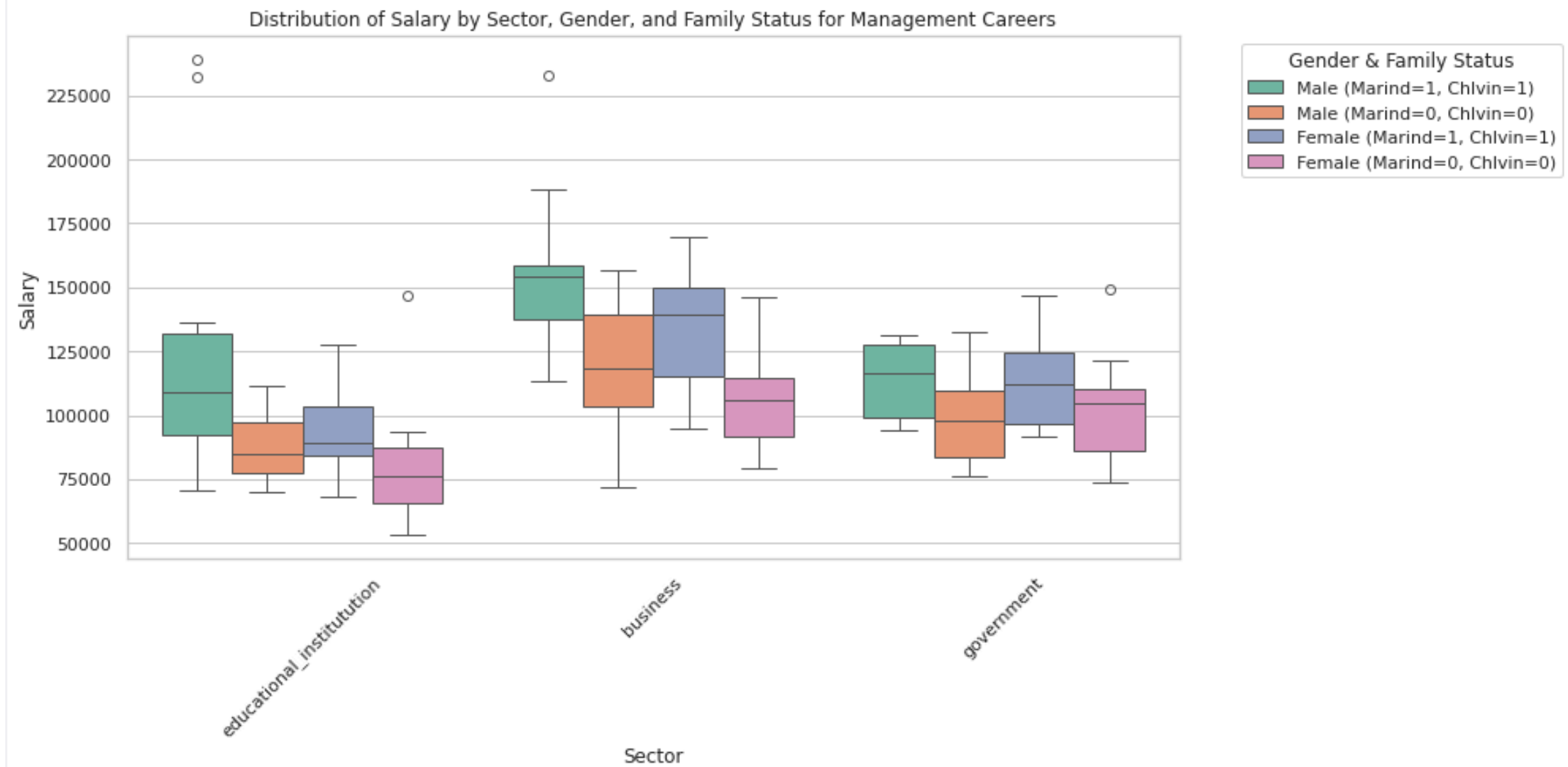
Percent Female in Management Roles by Sector



- **Education sector** has the highest proportion of females in management roles, particularly among unmarried and childless individuals. They have slightly higher female representation (55%) compared to those married with children (48%).
- **Business and government sectors** have lower female representation, with married individuals with children experiencing the lowest representation.
- Visualization supports the alternative hypothesis that the traditional family structures limit women's access and representation rate to management roles.

Hypothesis 1

Distribution of Salary by Sector, Gender, and Family Status



- **Business Sector:** Men in traditional family structures have the highest median salary. Women in the same group earn less despite wider salary variability.
- **Government Sector:** Minimal wage gap in traditional family structures, with similar median salaries for men and women. Unmarried individuals earn lower salaries but face a smaller gender gap.
- **Educational Institutions:** Men in traditional family structures earn the most, while women in the same group earn more than unmarried women but still less than men.
- Traditional family structures widen the gender wage gap across most sectors, except in government. Unmarried individuals earn the lowest salaries overall.

Chi-Square Test

- **Chi-Squared value (6.98)** : indicates a greater deviation from the null hypothesis.
- **P-value (0.0306)**: less than $\alpha = 0.05$. Statistically significant relationship between family status and female representation.
- Reject the null hypothesis.
- Evidence of a significant relationship between family status and female representation in management roles across sectors.

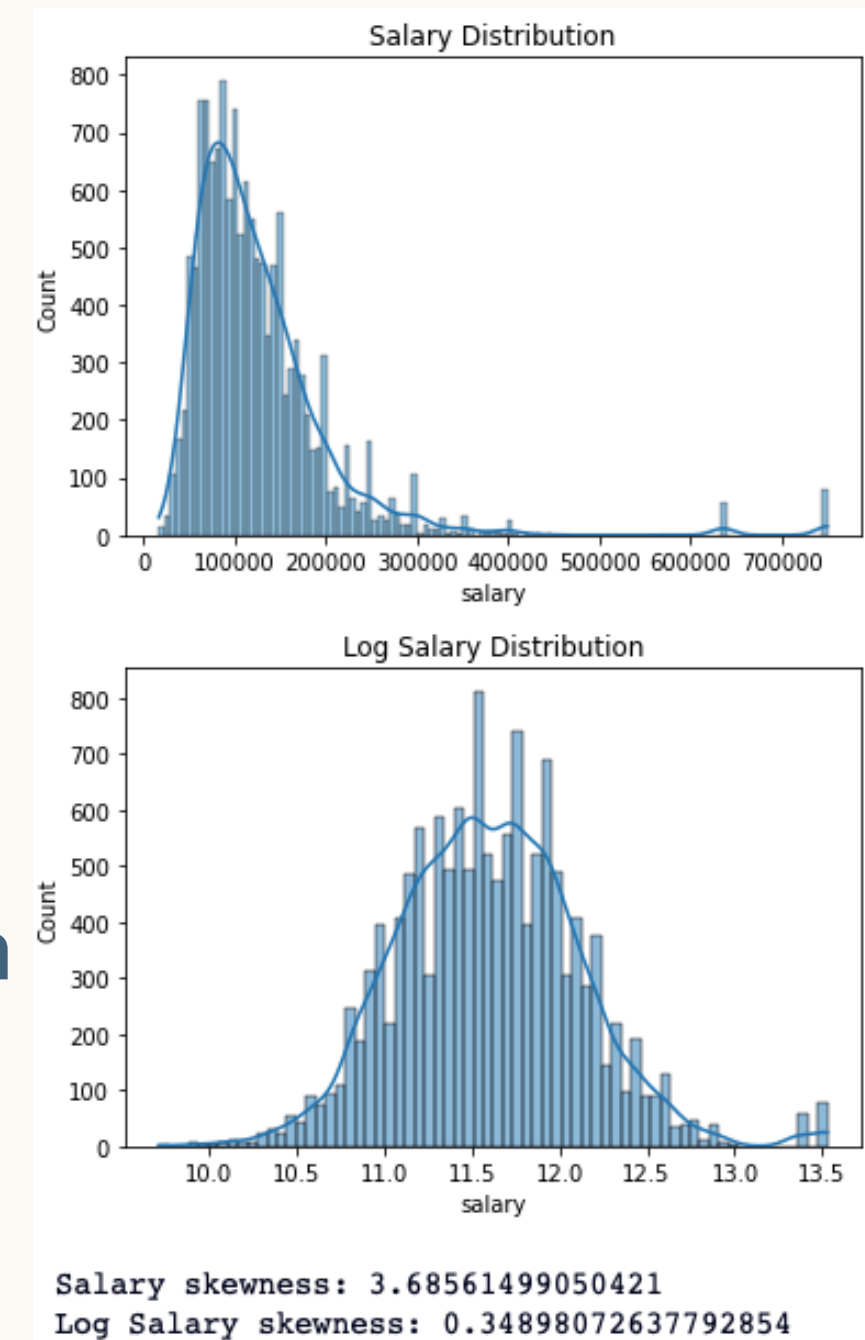
Chi-Squared Test Results
Chi-Squared Value: 6.98
p-value: 0.0306
Degrees of Freedom: 2

Control Variables:

- **Children's Age Groups:** intensifies the relationship between family structure and gender representation, emphasizing the dual burden women face
- **Employer Size:** large employers show stronger effects, smaller employers show weaker effects.
- **Hours Worked:** significant impact observed in 60–70 hr. range, typical of high-level management roles.
- **Age Groups:** greatest impact during mid-career stages (35–50 years) due to peak caregiving responsibilities.
- **Established Businesses:** traditional workplace norms in established businesses increase gender biases compared to newer organizations who show less bias.

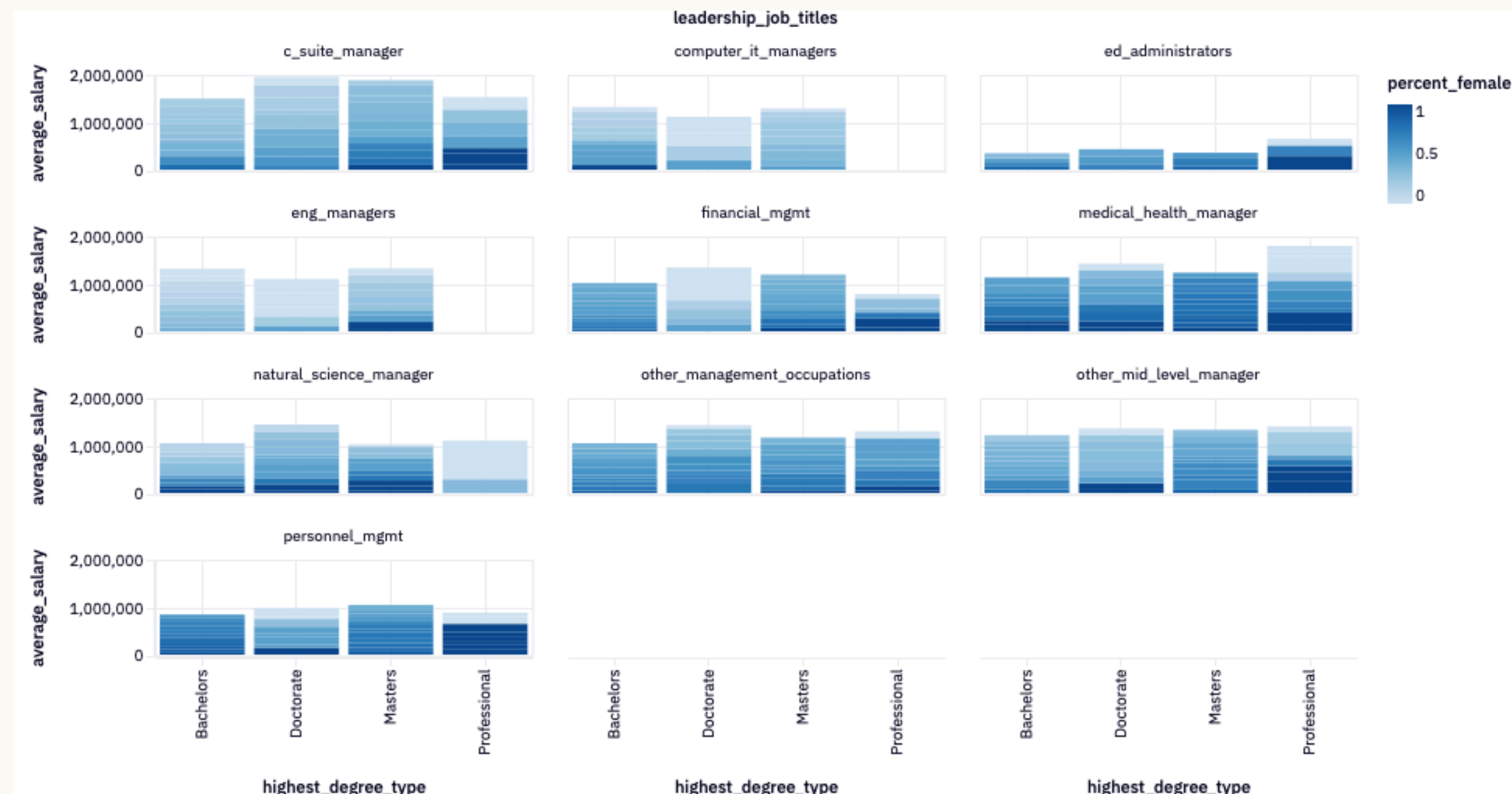
Multiple Regression

- The original data is highly skewed (3.69) due to outliers, distorting results. Log transformation reduces skewness (0.35), ensuring valid and reliable regression analyses.
- Women earn 8.32% less than men on average (significant).
- Married individuals earn 21.26% more, but marriage widens the gender wage gap by 7.14% (significant).
- Children increase salary by 7.23% but have no significant effect on the gender wage gap ($p=0.374$).
- Women are underrepresented in higher-paying roles and face significant pay gaps, especially in IT and education.
- Lower-paying sectors (e.g., education -26.90%) disproportionately affect women.
- Marriage amplifies the wage gap, while children do not. Family structure, particularly marriage, significantly impacts the gender wage gap in management roles.



EDA for Degree Types

- Relationship between degree type, average salary, and female representation in leadership roles within the business sector
- Higher education levels correspond to higher salaries
- Medical Health, Personnel, and Educational Administrators have higher proportions of women
- C-Suite, IT, and Engineering Managers have fewer women
- The present salary variation shows disparities contributing to the gender wage gap.

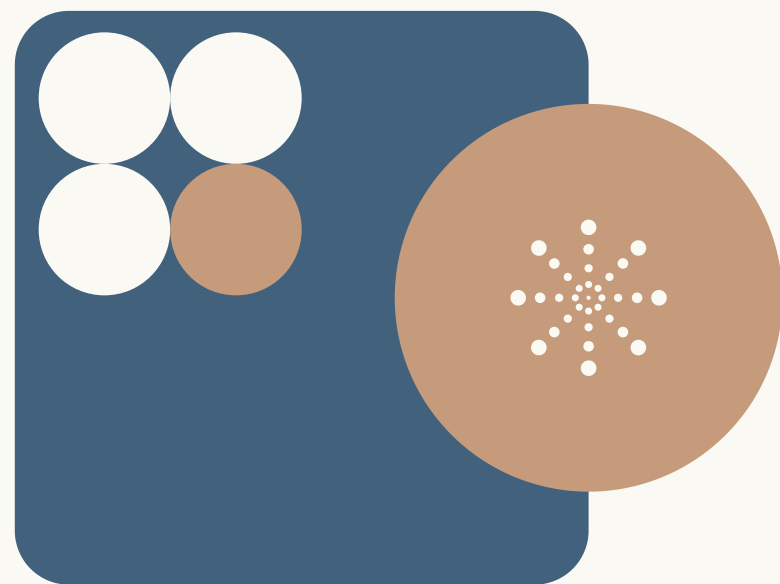


Weighted Least Squares and Mediation Analysis

- WLS: Women with professional degrees in management roles in business sector have largest salary premium at 14.34% lower than reference group
- WLS: Women with PhDs get 6.54% higher salary, and master's 1.79% lower than reference group
- All p-values significant.
- Education Level as a Mediator: Percentage of the gender gap explained by mediators is 30.13%.
 - Accounts for about one-third of observed gender wage gap in management roles in business sector
- Reject the null hypothesis based on these two tests



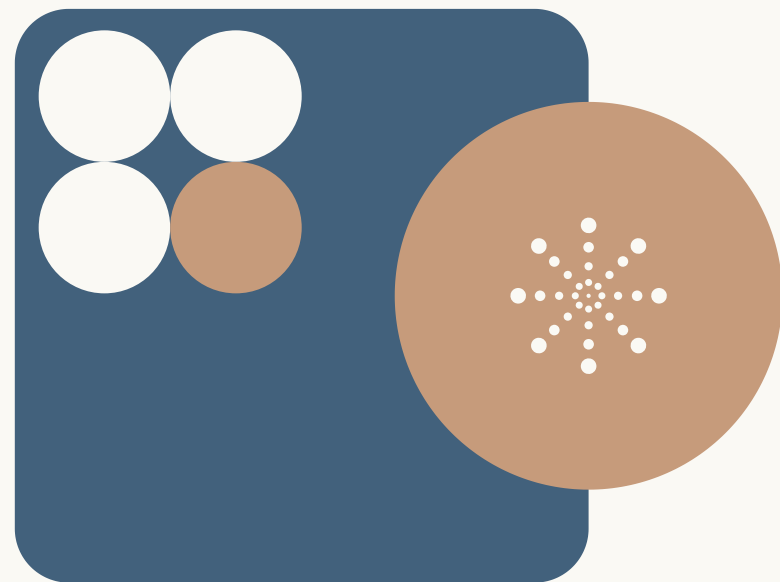
Conclusions



01. Hypothesis 1 (reject the null). Traditional family structures shape the gender wage gap and representation in management. Married men with children earn more, especially in business, while women in similar roles face wage gaps and underrepresentation. Unmarried individuals, particularly women, earn less across sectors. Regression results show marriage impacts the gender wage gap, emphasizing the need for interventions.

02. Hypothesis 2 (reject the null). Weighted least squares and mediation analyses show a significant link between gender, degree type, and the wage gap. Education and occupational segregation explain part of the gap, at 30.13%. Salary effects vary by degree type, particularly for women.

Policy Recommendations



01.

Hypothesis 1

Organizations should ensure gender and family structure have minimal impact on career outcomes, including conducting pay equity audits, establishing transparent promotion criteria, and providing training to mitigate unconscious bias.

02.

Hypothesis 2

Our analysis shows women with professional degrees earn 14.34% less than the reference group. Private employers, such as law firms and medical practices, should undergo pay audits, and stronger policies should ensure equal pay for equal qualifications.

03.

Broad Recommendation

Companies should report salary statistics by gender, role, education, bonuses, and promotions to increase accountability and address disparities. Similar policies already exist in other countries.